Recognition paths: when does a recognizer respect logical identity, and how would we know?

Working draft (2026-09-06). All sections written; the Qwen2.5-1.5B row of Section 4.3 is pending its run. Labels follow `AGENTS.md`: PROVED, OBSERVED, HYPOTHESIS, OPEN, CONFOUND. Source and data: github.com/Kairose-master/ {recognition-paths, proof-path-invariance}; huggingface.co/jinu0633.

Abstract (draft)

Whether a language model "respects logic" has no answer until three things are fixed: which logical identity, which behavioral identity, and which finite family of tests decides between them. We fix all three. Logical identity is theory-level Horn entailment on ordered premise traces; behavioral identity is two-sided contextual indistinguishability at the answer position; both are congruences on the same free monoid, and the model respects logic exactly when the first refines the second. We prove in Lean, without Mathlib, that this refinement is equivalent to the existence of a unique canonical monoid morphism from the logical quotient to the behavioral one; that premise reordering, repetition, and derivable-clause extension are consequences of that single inclusion rather than independent test relations; that a finite test family decides the question exactly when the identity it induces is closed under one-symbol extension, a closure failure naming the next test; and that the observation table's biextensional collapse is the identifiable object. We then build an instrument: frozen tables of prefix traces against continuation–query tests, read through order relations on the answer logits so that identity is exact, with preregistered statistics, a non-reading validity control, and a surface-controlled contrast between a maximal logic-preserving rewrite and a minimal logic-changing one. On three base and instruction-tuned models up to 1.5B parameters, no two logically identical traces are ever behaviorally identical; models that read the task are moved as much by permuting or redundantly extending the premises as by reversing one arrow. A recognizer built to be permutation-invariant by architecture and trained to 99% accuracy is exactly the syntactic quotient and only partly the semantic one. Reading the task, permutation invariance, and logical invariance are three distinct properties, in that order of difficulty; neither accuracy nor architectural invariance supplies the third.

1. Introduction

The question. Language models are sensitive to the order of premises (Chen et al. 2024), inconsistent across logically equivalent reformulations (LGMT 2026; CRTBench 2026), and their accuracy and their consistency come apart. These findings share a form: a set of inputs that a logic declares identical, and a model that treats them differently. What is missing is a statement of the object being tested. Which identity on inputs? Which identity on outputs? And since only finitely many inputs are ever tried, when has the test family decided the question rather than sampled it?

What we do. We give the three definitions, prove what follows from them, build an instrument that measures exactly what the definitions name, and use it on measured and constructed recognizers.

Contributions. (i) A machine-checked framework in which "respects logical identity" is the refinement of one congruence by another on a free monoid, with the equational consequences, the finite-test identification criterion, and the collapse of the observation table all proved (`recognition-paths`, Lean 4 core, no Mathlib). (ii) A readout of answer logits through order relations only, threshold-free and offset-invariant, which makes identity exact and immune to answer bias, together with counts of exactly identical profiles and collapse sizes. (iii) A surface-controlled statistic, permutation against single-arrow flip, with a non-reading validity control and a synthetic calibration. (iv) Constructed recognizers that realise the syntactic half of logical identity by architecture, on which the semantic half is measured in isolation.

What we do not claim. Order sensitivity and accuracy–consistency gaps are known; our small-model findings replicate them. The mathematics is classical (Myhill–Nerode, Angluin, Chu spaces); we assemble it and check it. Nothing here concerns frontier models.

2. Framework (PROVED)

2.1 Traces, queries, logical identity

Fix a type of atoms. A Horn clause is a pair of atom lists (body, head); a trace is a list of clauses; a query is a list of hypothesis atoms and a goal atom. The theory of a trace, $\Gamma(w)$, is its set of clauses. Entailment is semantic: every valuation modelling $\Gamma(w)$ and the hypotheses makes the goal true. Two traces are logically identical, $u \equiv_L v$, when they entail the same queries. This is an equivalence and a two-sided congruence for concatenation; since Γ forgets order and repetition it is commutative and idempotent, so $L = \Sigma^*/{\equiv_L}$ is a commutative idempotent monoid. Permutations are logically identical; appending a derivable clause is logically invisible (`Horn.lean`).

2.2 Recognizers and behavioral identity

A recognizer is $\rho : \Sigma^* \times Q \to O$. Right-context identity $u \equiv_\rho v$ asks that no continuation and query separate u from v; two-sided identity $u \approx_\rho v$ asks the same in every left and right context. The latter is a congruence, so $B = \Sigma^*/{\approx_\rho}$ is a monoid. The Nerode quotient $\Sigma^*/{\equiv_\rho}$ is the extensional collapse of the prefix realization (states = traces, tests = continuation– query pairs), and B acts on it as its transition monoid; every state is reached from the empty trace (`Recognition.lean`, `Nerode.lean`).

2.3 The factorization theorem and its consequences

${\equiv_L} \subseteq {\approx_\rho}$ holds iff there is a unique representative-preserving $F : L \to B$; F is then a monoid morphism. The reverse inclusion gives $G : B \to L$; both give $L \simeq B$. Under ${\equiv_L} \subseteq {\approx_\rho}$, swapping two blocks of premises, repeating a block, permuting a block, and appending a derivable clause are all behaviorally invisible in every context. In the vocabulary of metamorphic testing: reordering, duplication, and redundancy relations are not independent; they are theorems of one inclusion (`Factorization.lean`, `Recognition.lean`).

2.4 When a finite test family has decided

Let T be a family of tests (z, q) containing the direct queries ([], q). The identity it induces, $\equiv_{\{\rho,T\}}$*, equals* $\equiv_\rho$ *if and only if it is closed:* $u \equiv_{\{\rho,T\}} v$ *implies* $ua \equiv_{\{\rho,T\}} va$ for every symbol a. When T is not closed there exist $u \equiv_{\{\rho,T\}} v$ and a test $(a \cdot z, q)$ with $(z, q) \in T$ separating them, so the criterion is constructive. For length-bounded families, one more level of continuation is one more symbol of prefix, and a stable level identifies $\equiv_\rho$. Two-sided versions hold for $\approx_\rho$ (`Identification.lean`). The counting bound (n Nerode classes force stabilisation by length $n-1$) is standard and not formalised (OPEN).

2.5 The table as a Chu space

Quotienting tests by equal state profiles, dually to the state collapse, gives the biextensional collapse, in which tests separate states and states separate tests; silent extensions do not change it (`Biextensional.lean`). The counts "distinct rows" and "distinct columns" of an observation table are the sizes of these quotients. Under the broadest reading of "geometry", this point–test duality is the geometry that cannot be removed; metric structure is a further choice.

2.6 Specifications for constructed recognizers

A theory-factoring recognizer $\rho(w, q) = f(\Gamma(w), q)$ is invariant under permutation and repetition of blocks by construction; the ideal recognizer $\rho(w, q) = \mathrm{Entails}(\Gamma(w), q)$ is logically invariant and recoverable, so $L \simeq B$. The gap between them, identification of consequence-equivalent but syntactically different clause sets, is what training must supply (`Specification.lean`).

3. Instrument

3.1 Tables

Eight Horn logical classes over five atoms, chosen so that the class-level gold matrix has full rank and half of its cells are positive. Rows are prefix traces (serializations, and in later tables doubled prefixes, single-arrow flips, derivable-clause extensions); columns are continuation–query pairs. Four frozen tables (`hankel_v0`–`v3`), each with a generator, an independent validator that re-derives every gold label, a SHA-256 lock, and CI. Row equivalences inside a class are Lean-certified; cell gold labels are forward-chaining, not Lean-certified per cell.

3.2 Readout

Each cell records the logit pair (positive candidate, negative candidate) at the answer position, one forward pass, float32, no sampling. The analysis reads only order relations: the decision [pos > neg] and, for two queries after the same continuation, the preference [margin(q1) > margin(q2)]. This is threshold-free and invariant to the common logit offset, which had made the raw metric table rank one, and it is immune to the answer-bias saturation that discrete readouts inherit. Distances are Hamming counts of separating tests: the number of columns of the Chu space that distinguish two rows.

3.3 Statistics

All preregistered before the corresponding runs, with predictions recorded. Gate R: comparative accuracy on query pairs whose gold labels differ. Gate I: within-class median Hamming over between-class median. S: median over classes of d_perm / d_flip on the surface-controlled table. T, U, V: on the semantic-rewrite table, derivable extension against flip, length-matched swap against flip, and the same on columns where gold ties so that accuracy imposes nothing. E: counts of exactly identical profiles. A non-reading recognizer (Pythia-70M) is the validity control; a synthetic gold-plus-noise recognizer is the calibration showing that T and U are passed by accuracy alone while V is not.

3.4 Three corrections

The instrument corrected itself three times, each recorded. A Euclidean readout was dominated by a logit offset (rank one) and by scale differences across renderers; the Boolean readout replaced it. Gate I, valid against the control, was found to reward shared surface content: the class pair differing only in the direction of all three arrows was closer than a class to its own serializations; the surface-controlled contrast replaced it as the primary statistic. The apparent invisibility of repeated continuations was found in the control at the same rate and downgraded to a property of the prompts.

4. Measured recognizers (OBSERVED)

Recognizers: Pythia-70M and Pythia-410M (base, `step143000`), Qwen2.5-0.5B-Instruct and Qwen2.5-1.5B-Instruct (raw prompts, no chat template), one forward pass per prompt in float32 on CPU, both answer logits recorded.

4.1 Reading is not identifying

On the Boolean table (`hankel_v1`), Gate R and Gate I with their controls:

recognizer	comparative acc. (Gate R > 0.55)	within/between median Hamming (Gate I < 1)
Pythia-70M	0.456	1.42
Qwen2.5-0.5B	0.867	0.561

Qwen reads the task and passes Gate I; the control fails both, so the gate discriminates. But no recognizer has a single logical class whose serializations have identical profiles (0 of 8 for every recognizer, all 40 rows distinct), and under a matched between-class statistic the control also falls below 1. Gate I rewards shared surface content: for Qwen the class pair differing only in the direction of all three arrows is closer than a class to its own serializations. This is CRTBench's accuracy– consistency gap at small scale, with the addition that the readout is exact: identity never occurs.

4.2 Surface outweighs logic, and scale moves it

On the surface-controlled table (`hankel_v2`), `S` = median over classes of d_perm / d_flip:

recognizer	`S` pooled	bullets	prose	comparative acc.
Pythia-70M (control)	0.97	1.20	1.11	0.48
Qwen2.5-0.5B	1.27	0.75	1.80	0.84
Qwen2.5-1.5B	0.99	0.80	1.25	0.89

The statistic is one-sided: an accurate recognizer with no invariance reaches 0.5–0.6 because flips change gold decisions it tracks. So the diagnostic values are the ones above 1: for the 0.5B reader, and under the prose renderer at both scales, reordering the same three premises moves behavior more than reversing one arrow. From 0.5B to 1.5B the pooled value falls from 1.27 to 0.99 (HYPOTHESIS: logic rises over surface with scale within the family; two points).

4.3 Semantic rewrites are not identified

On the semantic-rewrite table (`hankel_v3`), a derivable-clause extension (Lean-certified logically invisible) against a flip:

recognizer	`T` red/flip	`V` free columns	`E` exact identities
Pythia-70M (control)	1.00	1.50	0 / 14
Qwen2.5-0.5B	1.11	1.50	0 / 14
Qwen2.5-1.5B	⟨pending⟩	⟨pending⟩	⟨pending⟩

For the 0.5B reader a logically invisible extension moves behavior as much as a logical change, and more where accuracy imposes nothing (`V` 1.5, equal to the control). The behavioral quotient does not contain the consequence relation at any level the table sees.

4.4 Closure, idempotence, and what is generic

The depth-one test family is not closed for any measured recognizer; for Qwen the witness pair is logically identical and separated only at depth two, the situation the identification theorem describes. Repeated continuations agree with their single versions on about 95% of tests for reader and control alike, so this is a property of the prompts, not of logic. The largest single source of within-class distance is a first-premise primacy effect on the query mentioned first (CONFOUND), and the atom-label family changes the metric invariance defect by about 3× (CONFOUND). Premise-by-premise margin increments are right-skewed, heavy-tailed, drift toward the positive answer, do not accumulate with depth, and shrink with level: not a diffusion (EXPLORATORY).

5. Constructed recognizers

Three recognizers trained on synthetic Horn traces with the eight evaluation classes held out up to atom relabelling, 6000 steps, batch 128: `set` (max-pooled clause encodings: permutation- and repetition-invariant by construction, ~170k parameters), `seq_aug` (causal transformer with permutation/repetition augmentation, ~100k) and `seq_fixed` (the same without augmentation). Evaluated on the same tables through the same statistics; one rendering, so no renderer split.

5.1 The syntactic half, exactly

The `set` recognizer is, on `hankel_v1`, exactly the syntactic quotient: its 40 rows collapse to the 8 logical classes, all eight classes have identical serialization profiles, `S` = 0 on `hankel_v2`, and the depth-one family is closed. Both seeds. No causal recognizer, at any of four seeds or at three times the budget (validation accuracy 0.98), has one class with identical profiles or fewer than 39 distinct rows. Augmentation buys approximate permutation invariance (`S` 0.49 at convergence); architecture buys exact invariance; accuracy buys neither.

5.2 The semantic half, partly

On `hankel_v3` the `set` recognizer, 99% accurate on classes it never saw, identifies none of the 14 derivable-clause extensions exactly in one seed and two in the other; on the columns where accuracy imposes nothing it moves 0.62–0.80 times as far under a semantic rewrite as under a logical flip. Every LLM, the control, and seven of eight causal toy runs have `V` above 1 (medians 1.5–2.3). So the architecturally invariant recognizer is the only one in which a semantic rewrite is detectably less disruptive than a logical change, and even there most of the semantic half is missing and almost none of it is exact. (Two seeds; suggestive.)

5.3 What the constructed recognizers add

They separate three properties that the LLM measurements conflate: reading the task (all accurate recognizers), permutation invariance (exact only by architecture, approximate by augmentation), and logical invariance (present partly, and only where the syntactic half was exact). They also show that `S < 1` and Gate I are reached by accuracy and surface alone, which fixes the reading of Section 4.

6. Discussion

Three properties, ordered. Reading the task, respecting the syntactic part of logical identity, and respecting its semantic part are distinct, and every recognizer we measured or built sits on that ladder at a different rung. Accuracy does not climb it; architecture climbs one rung exactly; nothing we tried climbs the last.

What a finite table can conclude. By the identification theorem, a test family decides `≡_ρ` when it is closed. On the measured recognizers the depth-one family is not closed, and the theorem names the column a deeper table needs. On the `set` recognizer the same family is closed, which is the theorem working as intended: the recognizer's quotient is coarse enough for the family to capture it.

Exact versus approximate. A statistical recognizer never produces two identical profiles for logically identical inputs; the exceptions here are forced by construction. The right long-run object is a canonical distance with a certified tolerance, not an equality (OPEN).

Metric and duality. Metric geometry on the logits misled twice (offset, scale); the Boolean readout keeps only the point–test duality of the Chu space, and every exact statement in this paper lives there.

Limitations. One Horn fragment, five atoms, eight classes; models up to 1.5B without chat templates; no sampling; toy constructed models at one or four seeds; gold labels by forward chaining, not certified per cell.

Related work and novelty. `RELATED_WORK.md`, `NOVELTY.md`: the empirical phenomena are known; the framework, the exact readout, the one-sided surface statistic with its control, and the separation of permutation from logical invariance are, on the texts read, not.